NEXORA

Day 1 Requirements Specification Document

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1.1 Executive Summary

Nexora is a centralized mentorship and career-growth platform designed to connect students and trainees with mentors, trainers, and organizations through structured, bookable, one-on-one interactions. The MVP release scopes a single end-to-end workflow—One-on-One Mentor Slot Booking—built strictly to production standards rather than as a proof of concept. This ensures that the core authentication, data, and notification foundations can scale seamlessly without architectural rework into future modules like internship tracking, assignment management, and AI-guided career features.

1.2 Problem Statement & Industry Context

Early-career trainees and students consistently face three structural bottlenecks:

• Informal Discovery: Mentorship access is currently unscalable, relying heavily on cold DMs or ad-hoc introductions with no searchable directory mapped to explicit skills or real availability.

• Scheduling Friction: Coordinating a simple session results in multi-message back-and-forth loops across fragmented time zones, lacking a single canonical source of truth.

• Lack of Accountability: Post-session workflows lack continuity, structured records of action items, or administrative oversight mechanisms to track actual engagement.

1.3 Existing System Analysis vs. Nexora Architecture

Existing Tool	What It Solves	Gap For Nexora's Use Case
Calendly / Cal.com	Single-user availability scheduling and booking links.	No concept of a role model triangle, no organization-level oversight, and no built-in program cohort admin tooling.
LinkedIn	Discovery and informal outreach to professional networks.	No structured booking ledger, low response rates, and zero capability to verify real availability.
Discord / Slack	Real-time community discussion and ad-hoc communication.	Mentorship remains unstructured and unscheduled, diluting mentor time across public channels with no 1:1 accountability.
Google Forms + Sheets	Manual program intake and tracking for student cohorts.	No real-time race-condition control, no conflict prevention, and completely unscalable without manual coordination.

1.4 Comprehensive Functional Requirements

- **Authentication & Identity**

ID	Requirement	Priority
FR-AUTH-01	Users can register with email + password, with email verification required before first login.	Must
FR-AUTH-02	Users can authenticate via Supabase Auth (or Firebase Auth) supporting email/password and OAuth (Google) sign-in.	Must
FR-AUTH-03	Users select a role (Student, Mentor) at signup; Admin roles are provisioned only by an existing Admin.	Must
FR-AUTH-04	Passwords must meet a minimum complexity policy and are never stored or logged in plaintext.	Must
FR-AUTH-05	Users can request a password reset via a time-limited, single-use email link.	Must
FR-AUTH-06	The system uses Supabase Auth (or Firebase Auth) for secure session management and user authentication.	Must
FR-AUTH-07	Users can securely log out and terminate their active session.	Must
FR-AUTH-08	Future Scope: Multi-factor authentication may be introduced for enhanced account security.	Should

- **Student / Trainee**

ID	Requirement	Priority
FR-STU-01	A student can create and edit a profile (name, bio, skills, interests, college/organization, resume link).	Must
FR-STU-02	A student can browse a searchable, filterable directory of mentors (by skill, domain, availability, rating).	Must
FR-STU-03	A student can view a mentor's public profile, including bio, expertise tags, and aggregate rating.	Must
FR-STU-04	A student can view a mentor's real-time available slots in their own local timezone.	Must
FR-STU-05	A student can book an available slot, attaching an optional topic/goal note.	Must

FR-STU-06	A student can view all upcoming and past bookings in a personal dashboard.	Must
FR-STU-07	A student can cancel or request to reschedule a booking within the policy-defined window.	Must
FR-STU-08	A student receives confirmation, reminder, and cancellation notifications via email and in-app.	Must
FR-STU-09	A student can leave a rating and optional written feedback after a completed session.	Should
FR-STU-10	A student can save/favorite mentors for quick re-booking.	Could

- **Mentor / Trainer**

ID	Requirement	Priority
FR-MEN-01	A mentor can create and edit a public profile (bio, expertise tags, experience, organization, LinkedIn/portfolio link).	Must
FR-MEN-02	A mentor's profile requires Admin verification before appearing in the public directory.	Must
FR-MEN-03	A mentor can define recurring weekly availability templates (e.g., Tuesdays 6-8 PM).	Must
FR-MEN-04	A mentor can define one-off availability slots outside the recurring template.	Must
FR-MEN-05	A mentor can block out exceptions (vacation, unavailability) that override recurring templates.	Must
FR-MEN-06	A mentor can view all upcoming bookings with mentee context (name, profile, attached note).	Must
FR-MEN-07	A mentor can cancel or propose a reschedule for a booking, triggering mentee notification.	Must
FR-MEN-08	A mentor can mark a session as Completed or No-Show after the scheduled time has passed.	Must
FR-MEN-09	A mentor can view their own session history and aggregate rating.	Should
FR-MEN-10	A mentor can set a maximum number of active bookings per week to control load.	Should

- **Administrator**

ID	Requirement	Priority
FR-ADM-01	An Admin can view, search, and filter all registered users (students and mentors).	Must
FR-ADM-02	An Admin can verify, suspend, or reinstate a mentor profile.	Must
FR-ADM-03	An Admin can view all bookings platform-wide with filters (status, date range, mentor, student).	Must
FR-ADM-04	An Admin can manually cancel or reassign a booking in dispute or error scenarios.	Must
FR-ADM-05	An Admin can provision additional Admin accounts with scoped permissions.	Must
FR-ADM-06	An Admin can view platform-wide metrics (active users, booking volume, completion rate).	Must
FR-ADM-07	An Admin can configure global policy settings (cancellation window, reminder timing).	Should
FR-ADM-08	Administrative actions are logged for monitoring and accountability.	Must

- **Booking Engine**

ID	Requirement	Priority
FR-BOOK-01	The system prevents double-booking of a single slot via atomic, transaction-safe slot locking.	Must
FR-BOOK-02	A booking transitions through a defined state machine: Pending -> Confirmed -> Completed / Cancelled / No-Show.	Must
FR-BOOK-03	Slot availability updates in real time across all connected clients via WebSocket push.	Must
FR-BOOK-04	Cancellations within the policy window automatically release the slot for re-booking.	Must
FR-BOOK-05	The system enforces a configurable minimum-notice window for new bookings (e.g., no bookings within 2 hours of the slot).	Should
FR-BOOK-06	The system supports timezone-aware slot display so mentor and student see correct local times.	Must

- **Notifications**

ID	Requirement	Priority
FR-NOTIF-01	The system sends a transactional email on booking confirmation, cancellation, and reschedule via the Resend Email API.	Must
FR-NOTIF-02	The system sends an automated reminder email/in-app notification a configurable interval before a session (default 24h and 1h).	Must
FR-NOTIF-03	In-app notifications are delivered in real time via WebSocket to connected clients and persist for offline users until read.	Must
FR-NOTIF-04	Users can configure notification preferences (email on/off per category).	Could

- **Reporting & Analytics (MVP Scope)**

ID	Requirement	Priority
FR-REP-01	The system records booking and completion events in a structured, queryable format suitable for future analytics expansion.	Must
FR-REP-02	Admins can export a CSV of bookings within a selected date range.	Should
FR-REP-03	The system computes and displays basic aggregate metrics (total bookings, completion rate, active mentors) on the Admin dashboard.	Must

- **Future Scope (Documented, Not Built in MVP)**

ID	Requirement	Priority
FR-FUT-01	Internship Application Tracking: students track multi-stage applications linked to organizations.	Future
FR-FUT-02	Assignment Management: mentors assign and grade structured tasks tied to mentees.	Future
FR-FUT-03	Event Registration: organizations publish and manage registration for workshops/webinars.	Future
FR-FUT-04	Learning Progress Tracking: longitudinal skill/competency tracking per student.	Future

FR-FUT-05	Community Discussions: threaded, moderated discussion spaces scoped by topic/cohort.	Future
FR-FUT-06	AI-Powered Career Guidance: LLM-driven recommendation engine for mentor matching and career paths.	Future
FR-FUT-07	Resume Review Workflows: structured, versioned resume feedback loop between student and mentor.	Future

1.5 Non-Functional Requirements (NFRs)

Category	Requirement Specification	Target Metric
Performance	API response latency for core read endpoints (directory browse, slot fetch) under standard load.	P95 < 300ms
Performance	Booking write execution (slot lock + reservation state commit) including notification trigger paths.	P95 < 800ms
Availability	Operational system baseline uptime, excluding planned maintenance windows.	99.5% Monthly
Scalability	Sustained transaction write load handling capacity at peak times (e.g., synchronized slot-release windows).	System should maintain stable performance under expected MVP usage.
Real-time Engine	WebSocket event pipeline distribution latency for slot-status modifications and notifications.	< 500ms
Data Durability	Persistent automated backup schedule constraints coupled with Point-in-Time Recovery capability.	Daily + 7-Day PITR
Usability	Fully responsive layout viewport adaptation boundaries.	320px - 1440px+

1.6 Security Requirements

ID	Security Control Specification	Priority
SEC-01	All traffic is served over HTTPS/TLS 1.2+; plain HTTP requests are forcefully redirected, never served.	Must
SEC-02	Authentication and session management are securely handled through Supabase Auth (or Firebase Auth) following industry best practices.	Must
SEC-03	Data ingestion layers strictly mandate parameterized queries or ORM mapping paradigms; raw SQL concatenation is forbidden.	Must

SEC-04	Role-based access control ensures users can only access authorized resources and data.	Must
SEC-05	All mutating network endpoints validate, sanitize, and unpack payload contexts server-side regardless of client checks.	Must
SEC-06	Passwords undergo one-way adaptive cryptographic hashing transformations (bcrypt/argon2) prior to persistence.	Must
SEC-07	Strict rate limiting constraints are applied across authentication routes to block brute-force or credential-stuffing exploits.	Must
SEC-08	File storage items are strictly scanned for size boundaries and signature vectors, then isolated within non-executable cloud locations.	Must
SEC-09	Important administrative actions are logged for monitoring and accountability.	Must
SEC-10	Production credentials and environment private vectors reside strictly within an isolated, monitored configurations engine.	Must
SEC-11	Cross-Origin Resource Sharing (CORS) rules strictly block access from unauthorized API domain pathways in production.	Must
SEC-13	Personally Identifiable Information (PII) uses column or disk encryption configurations at rest within the storage systems.	Must